

Hardware & Networking

C1Q26 Optical Market Update: Datacom Market Outlook Revised Higher; Positive for COHR, FN, and LITE

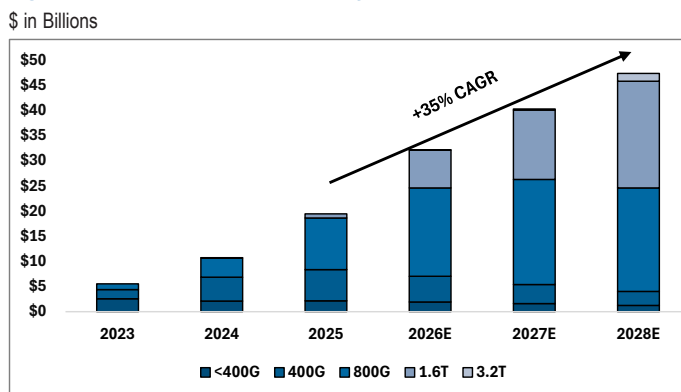
Industry analysts LightCounting have revised higher their optical forecast for the datacom market, which should largely be a positive for optical component suppliers in our coverage, including Coherent, Fabrinet and Lumentum; although, despite the raises, third-party analysts continue to highlight their conservatism in the forecasts, particularly starting in 2027, with limited capital spending visibility in their view from NeoClouds to support the growth trends in capex towards AI Infrastructure. *Key highlights from the latest forecast include:*

Growth Outlook Revised Higher For Datacom

The datacom growth outlook remains robust, with the total market forecast once again revised upward and now expected to expand at nearly a +35% CAGR through 2028, led primarily by 1.6T and, to a slightly lesser extent, 800G data rates. Additionally, the recent upward revisions to the outlook were also driven more by 1.6T, closely followed by revisions across 800G outlook. Importantly, attach rates of datacom solutions to XPU's continue to rise, with potential upside to both attach rates and the overall forecast, particularly in scale-up opportunities, such as multi-rack use cases in the near- to medium-term.

- **Datacom market is forecasted to expand at almost a +35% CAGR through 2028.** The latest outlook from LightCounting estimates that the datacom market will expand at a +35% CAGR from \$19 bn in 2025 to \$47 bn in 2028. From a data rate perspective, growth over the forecast period is being driven by 1.6T, which is expected to expand at ~200% CAGR to \$21 bn, followed by 800G, which is projected to grow at nearly +26% CAGR to \$21 bn. Additionally, 3.2T is expected to contribute modestly in 2028E, while 400G and lower data rates are expected to decline over the forecast period.

Figure 1: Datacom Market Forecast by Data Rate



Source: LightCounting.

- **Latest outlook embeds a sizable upward revision across both 1.6T and 800G market forecasts.** The latest outlook by LightCounting reflects an

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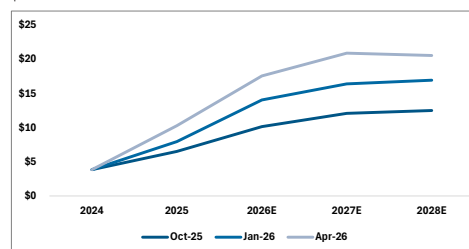
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average upward revision of +\$9 bn through the forecast period (average through 2028), driven primarily by a +\$5 bn increase in the 1.6T market forecast and a modestly lower +\$4 bn upward revision in the 800G market forecast, while other data rates in aggregate remained largely unchanged.

Figure 2: 800G Datacom Market Forecast

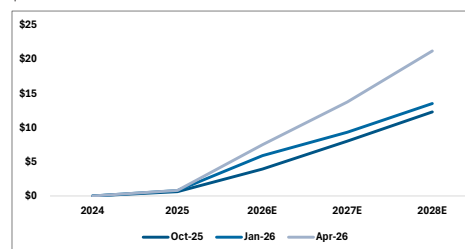
\$ in Billions



Source: LightCounting.

Figure 3: 1.6T Datacom Market Forecast

\$ in Billions

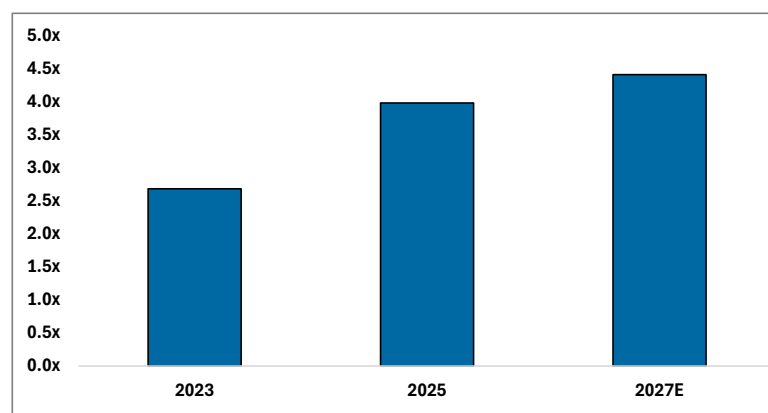


Source: LightCounting.

- Optical attach rate per XPU continues to expand.** As illustrated in Figure 4, based on the latest AI XPU forecast from J.P. Morgan Asia Technology Research, combined with LightCounting's datacom market outlook, we estimate that the blended attach rate for optical products tracked to over 2.5x per XPU in 2023, rising to a little under 4.0x in 2025 (vs. ~3.5x prior) and close to 4.5x in 2027 (vs. 4.2x prior), underscoring an increasing attach rate driven by the growth of larger cluster sizes that utilize more XPUs. That said, attach rates can vary by architecture, according to industry experts, with Google's use of optical circuit switches (OCS) resulting in an attach rate closer to 1.5x per TPU, while Nvidia deployments track closer to 6.0x per GPU.

Figure 4: Datacom Attach Rate

X



Source: LightCounting and J.P. Morgan Asia Technology team estimates.

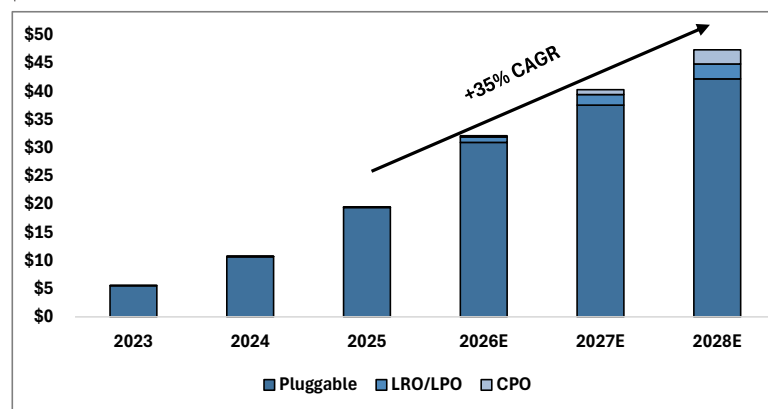
Expect Upside to Raised Industry Forecasts for CPO Market

Updated industry forecasts highlight that CPO market is expected to expand at a robust pace reaching \$2 bn by 2028 and expand further to ~\$11 bn by 2030. However, the latest guidance from Lumentum is already driving us to forecast their CPO revenues to be ~\$2 bn in CY28, implying overall CPO market TAM to be well ahead of \$2 bn level, which is currently being forecasted by the industry analysts for 2028. We believe that estimates from industry analysts are conservative and are likely to be revised higher in subsequent updates. Industry forecasts imply that the non-CPO market (i.e. transceiver market) is still expected to grow solidly over the period, expanding at a +20% CAGR and reaching \$45 bn+ by 2030. *Please see below for the latest outlook from industry analysts.*

- Majority of growth through 2028 is expected from pluggable transceivers while material contributions from CPO are expected only beyond 2028. Despite improving sentiment around the co-packaged optics (CPO) opportunity, the vast majority of datacom market growth over the next few years is expected to still be driven by traditional pluggable transceivers, which are projected to expand at ~30% CAGR to over \$40 bn in 2028. This is followed by LRO/LPO transceivers, which are expected to grow at a 380%+ CAGR to \$3 bn, while CPO is forecasted to expand at a 370%+ CAGR to \$3 bn.

Figure 5: Datacom Market Forecast by Technology

\$ in Billions

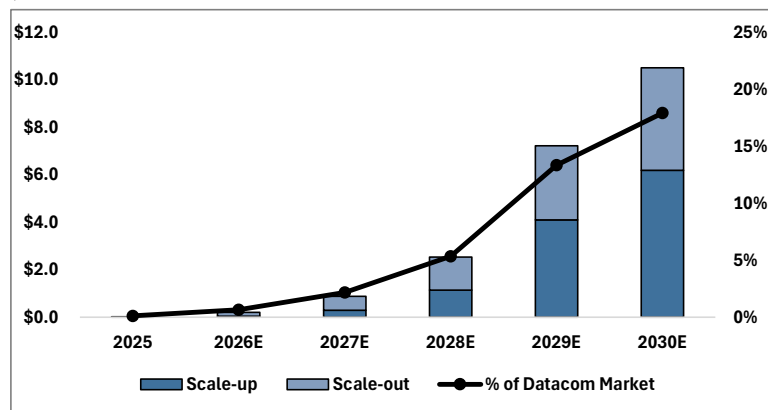


Source: LightCounting.

- CPO market to inflect in 2028 and beyond with scale-up use cases; Non-CPO market still expected to grow solidly. The CPO market is forecasted to grow from minimal revenue in 2025 to ~\$11 bn by 2030, with the real inflection point occurring in 2028 as scale-up use cases begin to ramp (*particularly dual- or multi-rack use cases*), with scale-up accounting for more than 45% of the CPO market from 2028 onward. Keep in mind, while CPO is expected to see sharp growth in the later years driven by scale-up deployments, it is still only expected to account for ~20% of the total datacom market by 2030, implying that the non-CPO market is expanding close to a +20% CAGR through the time period.

Figure 6: Long-term CPO Market Forecast

\$ in Billions



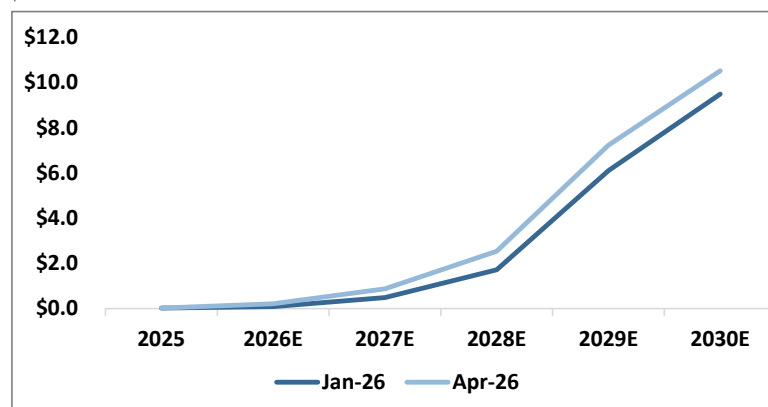
Source: LightCounting.

- Upward revision to outlook is driven primarily by scale-out CPO use cases. The

latest outlook for CPO by LightCounting reflects an average upward revision of ~\$1 bn in 2028 and beyond, relative to the prior update in January, driven almost entirely by upward revisions to forecasts for CPO adoption across scale-out use cases, while the outlook for CPO adoption across scale-up use cases remained largely unchanged.

Figure 7: Long-term CPO Forecast

\$ in Billions



Source: LightCounting.

- **Vast majority of 3.2T Ethernet ports are expected to be served via CPO by the end of the decade.** In terms of CPO penetration across different port speeds by the end of the decade, 1) 800G ports are expected to be overwhelmingly served by traditional transceivers (~80%), with CPO penetration expected to track close to 10%, 2) 1.6T ports are expected to be fairly balanced between traditional transceivers and CPO, and 3) a vast majority (~70%) of 3.2T ports are expected to be served by CPO.

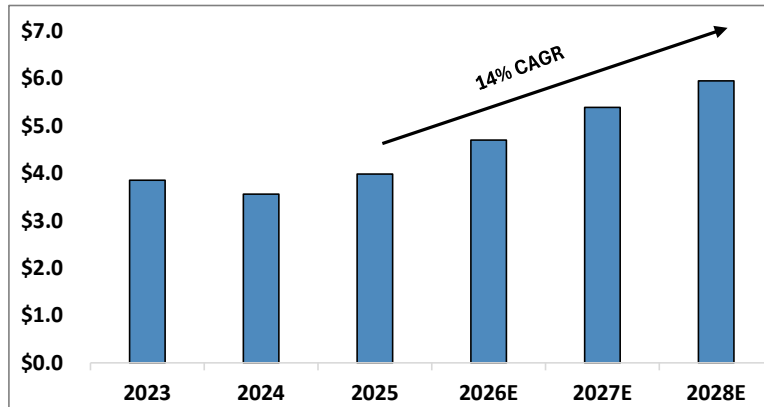
Telecom & DCI Growth Outlook Revised Modestly Lower

Telecom & DCI market growth outlook was lowered modestly, with the total market now forecasted to expand at +14% CAGR through 2028, driven by 800G and higher data rates. Downward revision to the outlook relative to prior forecast was driven primarily by a lowered outlook for high-speed optics (1.6T), across both ZR and Embedded form factors, where the market outlook is pushed out by a year or so as end-customers wait for availability of multi-rail amplifiers and multi-rail line systems.

- **Telecom & DCI market is forecasted to expand at +14% CAGR through 2028.** The latest outlook from LightCounting estimates that the Telecom & DCI market (CWDM and DWDM) will expand at a +14% CAGR from \$4 bn in 2025 to \$6 bn in 2028. From a data rate perspective, growth over the forecast period is being driven by 1.6T ramping up at over 130% CAGR reaching \$1 bn level by 2028, followed by 800G growing at 44% CAGR to \$2 bn+ over the same period.

Figure 8: Telecom & DCI Market Outlook

\$ in Billions

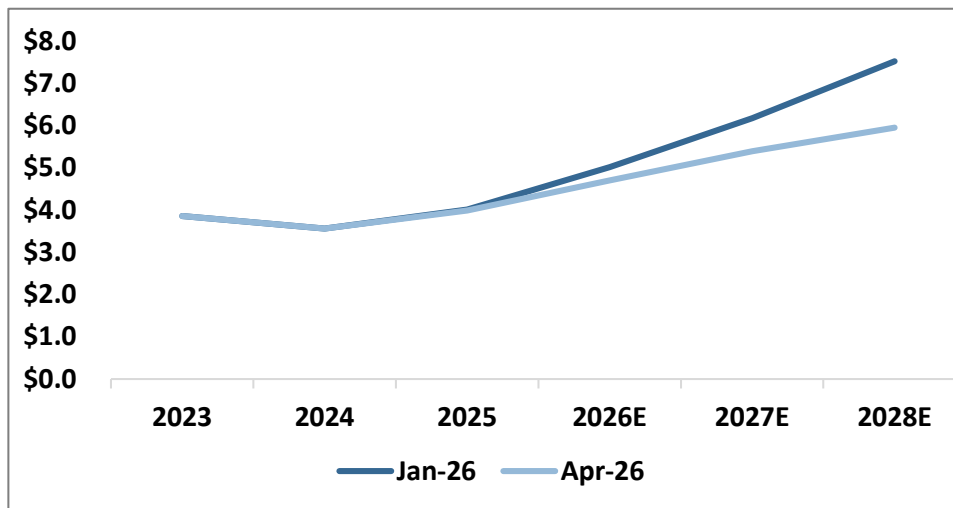


Source: LightCounting.

- **Latest outlook embeds a modest downward revision driven by 1.6T ZR/ZR+.** The latest outlook by LightCounting reflects an average downward revision of ~\$1 bn since the last forecast in January, driven primarily by a lowered outlook for high-speed optics (1.6T), across both ZR and Embedded form factors, where the market outlook is pushed out by a year or so as end-customers wait for availability of multi-rail amplifiers and multi-rail line systems.

Figure 9: Telecom & DCI Market Outlook

\$ in Billions



Source: LightCounting.

Robust Outlook for Optical Circuit Switching; Material Volumes Towards End-of-the Decade

The Optical Circuit Switching market is expected to expand from ~\$1.3 bn in 2026 to ~\$4.3 bn by 2031, at a CAGR of ~26%. Google is expected to remain a dominant player throughout the forecast period, and leverages 300-port and 128-port switches, for use cases like connecting up to 9k TPUs, and as a replacement for electrical spine switches. Key attributes driving adoption include high bandwidth, low latency, low power consumption, and the ability to future-proof networks by reusing switches across multiple interconnect generations.

Companies Discussed in This Report (all prices in this report as of market close on 14 May 2026, unless otherwise indicated)
Coherent Corp(COHR/\$404.94/OW), Fabrinet(FN/\$746.47/N), Lumentum(LITE/\$1,001.81/OW)

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Coherent Corp (COHR, COHR US) Price Chart



Date	Rating	Price (\$)	Price Target (\$)
19-Jul-23	OW	48.37	63
06-Feb-24	OW	49.42	70
07-May-24	OW	57.92	75
12-Aug-24	OW	63.34	85
04-Nov-24	OW	92.28	105
16-Jan-25	OW	96.45	110
05-Feb-25	OW	87.74	120
17-Apr-25	OW	55.45	80
08-May-25	OW	69.62	86
12-Jun-25	OW	81.51	100
17-Jul-25	OW	97.82	127
06-Nov-25	OW	134.63	180
05-Dec-25	OW	177.35	215
04-Feb-26	OW	229.18	245
09-Apr-26	OW	281.79	300
07-May-26	OW	344.67	380

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jul 29, 2019. All share prices are as of market close on the previous business day.

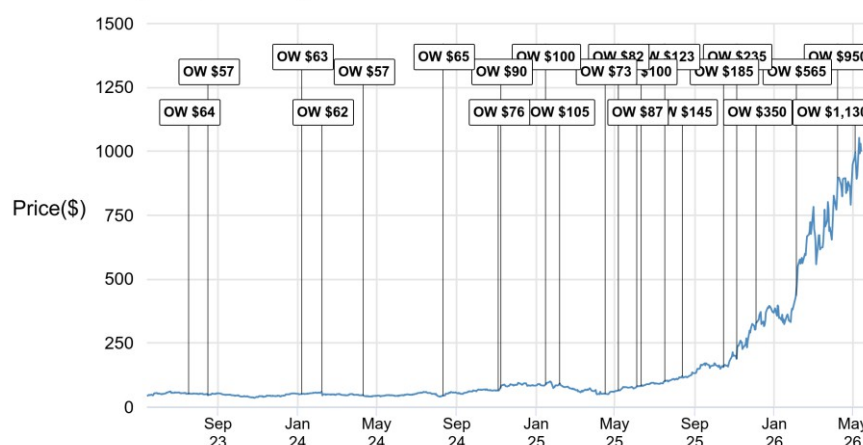
Fabrinet (FN, FN US) Price Chart



Date	Rating	Price (\$)	Price Target (\$)
19-Jul-23	OW	130.74	160
21-Aug-23	OW	117.85	174
06-Nov-23	OW	162.52	185
08-Jan-24	N	184.29	210
11-Apr-24	N	176.85	212
06-May-24	N	179.22	230
12-Aug-24	N	208.34	240
19-Aug-24	N	231.08	285
16-Jan-25	N	233.74	275
03-Feb-25	N	216.21	265
17-Apr-25	N	186.70	220
05-May-25	N	218.26	235
17-Jul-25	N	297.45	318
25-Aug-25	OW	294.35	345
16-Oct-25	OW	398.34	430
03-Nov-25	OW	440.57	530
16-Apr-26	N	685.81	700
04-May-26	N	706.53	680

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 09, 2011. All share prices are as of market close on the previous business day.

Lumentum (LITE, LITE US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends. Initiated coverage Aug 06, 2015. All share prices are as of market close on the previous business day.

Date	Rating	Price (\$)	Price Target (\$)
19-Jul-23	OW	52.10	64
17-Aug-23	OW	46.81	57
08-Jan-24	OW	50.47	63
08-Feb-24	OW	58.77	62
11-Apr-24	OW	45.34	57
12-Aug-24	OW	43.25	65
04-Nov-24	OW	65.35	76
08-Nov-24	OW	73.64	90
16-Jan-25	OW	88.40	100
06-Feb-25	OW	89.19	105
17-Apr-25	OW	52.21	73
07-May-25	OW	64.42	82
05-Jun-25	OW	80.28	87
12-Jun-25	OW	82.36	100
17-Jul-25	OW	99.63	123
13-Aug-25	OW	119.66	145
16-Oct-25	OW	156.57	185
05-Nov-25	OW	188.36	235
05-Dec-25	OW	327.85	350
04-Feb-26	OW	435.10	565
09-Apr-26	OW	896.02	950
06-May-26	OW	994.56	1,130

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